

ECE Probabilistic Systems Analysis

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Random Experiment Examples

Example 01: Select a ball from an urn containing balls numbered 1 to 50. Note the number of the ball

$$\{1, 2, \dots, 50\}$$

Example 02: Select a ball from an urn containing balls numbered 1 to 4. Suppose that balls 1 and 2 are black and that balls 3 and 4 are white. Note the number and color of the ball you select.

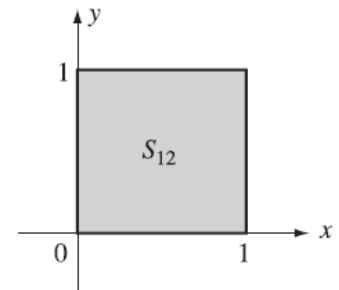
$$\{(1, b), (2, b), (3, w), (4, w)\} \quad \text{The ball is white and even-numbered} \quad \boxed{\{(4, w)\}}$$

Example 03: Toss a coin three times and note the sequence of heads and tails.

$$\{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\} \quad \text{The three tosses give the same outcome} \quad \boxed{\{HHH, TTT\}}$$

Example 04: Pick two numbers at random between zero and one.

$$\{(x, y) : 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1\}$$

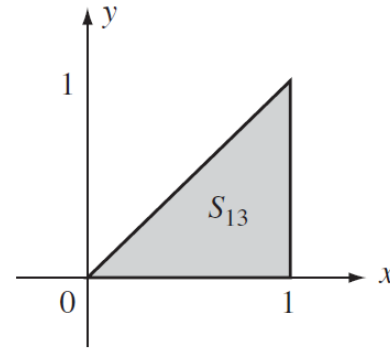


Example 05: A block of information is transmitted repeatedly over a noisy channel until an error-free block arrives at the receiver. Count the number of transmissions required.

$$\{1, 2, 3, \dots\}$$

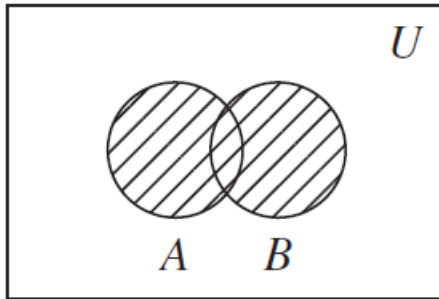
Example 06: Pick a number X at random between zero and one, then pick a number Y at random between zero and X

$$\{(x, y) : 0 \leq y \leq x \leq 1\}$$

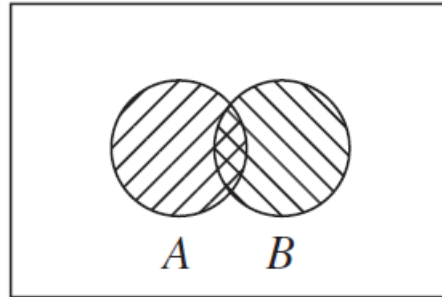


Venn diagram

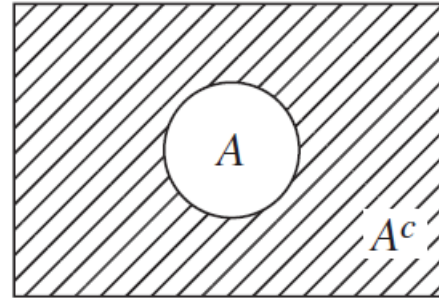
Union $A \cup B$



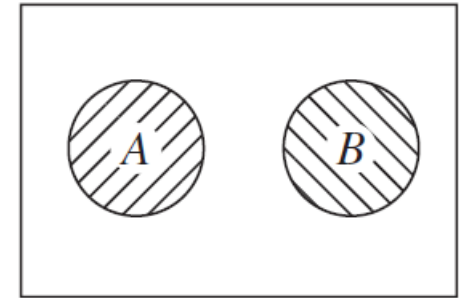
Intersection $A \cap B$



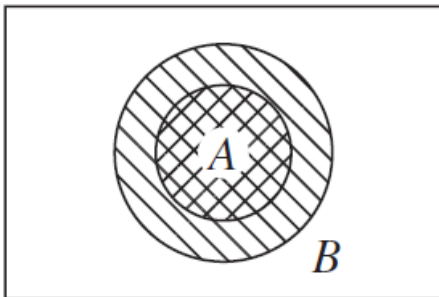
Complement A^c



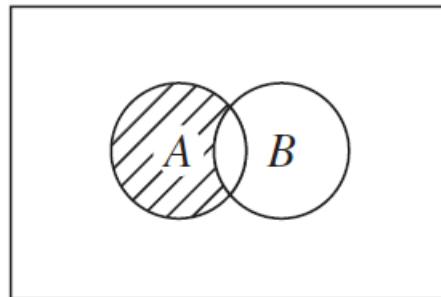
$A \cap B = \emptyset$



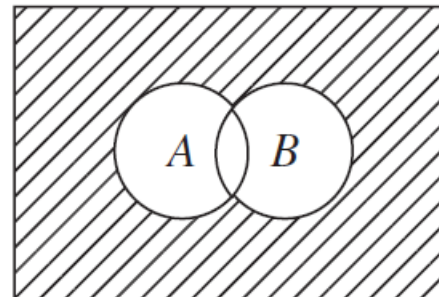
Subset $A \subset B$



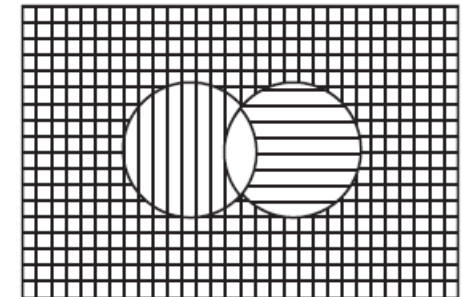
Relative Complement $A - B$



$(A \cup B)^c$



$A^c \cap B^c$



DeMorgan's rules

$$(A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (A \cap B)^c = A^c \cup B^c$$

Proof that $P[A \cup B] = P[A] + P[B] - P[A \cap B]$ If A and B are two joint events

Example 07:

Romeo and Juliet have a date at a given time, and each will arrive at the meeting place with a delay between 0 and 1 hour, with all pairs of delays being equally likely. The first to arrive will wait for 15 minutes and will leave if the other has not yet arrived. What is the probability that they will meet?

Example 08:

Out of the students in a class, 60% are geniuses, 70% love chocolate, and 40% fall into both categories. Determine the probability that a randomly selected student is neither a genius nor a chocolate lover.